

Selenium

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What's Selenium

- a portable **software testing framework** for **web applications**



Selenium	
Stable release	2.33 / May 22, 2013
Development status	Active
Written in	Java (programming language)
Operating system	Cross-platform
Type	software testing framework for web applications
License	Apache License 2.0
Website	seleniumhq.org

Selenium Components

- **Selenium IDE**
- **Selenium Client API**
- **Selenium Remote Control**
- **Selenium Grid**
- **Selenium WebDriver**

Selenium IDE

- A Firefox plug-in and is generally the most efficient way to develop test cases.
- Reference:
http://docs.seleniumhq.org/docs/02_selenium_ide.jsp#selenium-commands-selenese

Selenium Client API

- An alternative to writing tests in Selenese, tests can be written in various programming languages.
- With Selenium 2, a new Client API was introduced
- These tests then communicate with Selenium by calling methods in the Selenium Client API.

Selenium Remote Control

- Aka. **Selenium 1**
- The main Selenium project for a long time
- support for several languages and support for almost every browser out there.

Selenium Grid

- Allows you to run your tests in parallel, that is, different tests can be run at the same time on different remote machines.

Selenium Webdriver

- Selenium RC + Webdriver = **Selenium Webdriver** (aka. **Selenium 2**)
- WebDriver is designed to provide a simpler, more concise programming interface
- Reference:
http://docs.seleniumhq.org/docs/03_webdriver.jsp#introducing-webdriver

Installation

- **Official Documentation**
- http://www.seleniumhq.org/docs/03_webdriver.jsp
- **The 5 Minute Getting Started Guide**
- <https://code.google.com/p/selenium/wiki/GettingStarted>

Unit Testing Frameworks

- With Selenium 2, you can write tests in Java, C#, Ruby, and Python.
- Use your favorite unit testing tools
- The following uses JUnit

Instantiate a WebDriver

```
public class Selenium2Example1Test {  
    @Test  
    public void test() {  
        // Instantiate a webDriver implementation  
        WebDriver webdriver = new FirefoxDriver();  
    }  
}
```

- There's also:
 - ChromeDriver(),
 - OperaDriver(),
 - AndroidDriver(),
 - InternetExplorerDriver(),
 - Etc... (additional setting may needed)

Simple Test

```
public class Selenium2Example2Test {  
    @Test  
    public void test() {  
        System.setProperty("webdriver.chrome.driver",  
"src/main/resources/drivers/chrome/chromedriver-mac");  
  
        // Instantiate a webDriver implementation  
WebDriver webdriver = new ChromeDriver();  
  
webdriver.get(https://github.com);  
  
Assert.assertEquals("GitHub - Social Coding", webdriver.getTitle());  
    }  
}
```

Using the findElement() method

- `webdriver.findElement(By.id("header"));`
- `By.name("name")`
- `By.tagName("a")`
- `By.xpath("//a[@title='logo']")`
- `By.cssSelector(".feautures")`
- `By.linkText("Blog"))`
- `By.partialLinkText("Ruby")`
- `By.className("login")`

Element Events

- **WebElement**

- click()
- submit()
- sendKeys()
- clear()
- isDisplayed()
- isEnabled()
- etc...

Put them together

```
@Test  
public void searchResults() {  
    driver.findElement(searchBox).sendKeys(searchQuery);  
    driver.findElement(submitSearch).click();  
    String search = driver.getPageSource();  
    boolean searchResults = search.contains(searchQuery);  
    assertTrue(searchResults);  
}
```

Project 2 Requirements

- SUT requirements
- Test requirements
- Test-cases
- Implement test-cases with Selenium
- Makes a presentation

- Turn in the project 1 files